
BIOMASS ENERGY

1) What are the main types of OTEC power plants? Describe their working principle in brief with neat sketches?

OTEC:

OTEC or ocean thermal energy thermal conversion is a technology which converts solar radiation absorbed by the oceans to electric energy. The ocean's can be considered as the world's largest solar energy collector as it covers two third of the earth surface.

OCEAN TEMPRATURE DIFFRENCE:

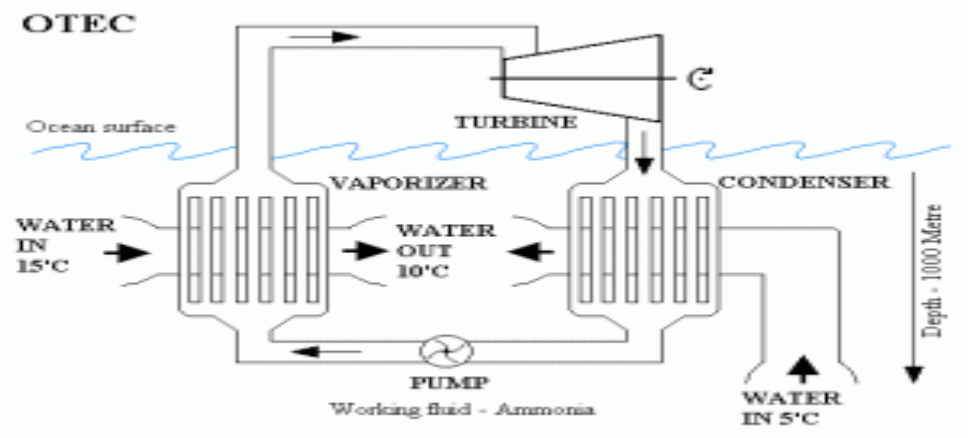
There is different temperature in the different layers of the oceans. This is because of the heat input from the sun at the surface of ocean. The surface at the top of the oceans are warmest and gradually the temperature decreases with in depth. But in the polar regions the temperature at the surface of ocean is low,so there is no gradual change in temperature.

PRINCIPLES OF OTEC PLANT OPERATION:

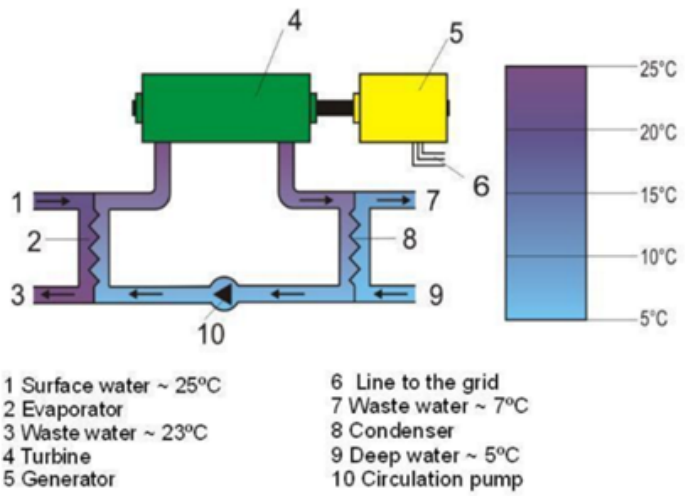
The working principle of an OTEC plant is that it uses the warm water to heat and vaporize a liquid (working fluid).And this working fluid develops pressure which forces it to evaporate.and the expanding vapour runs through a heat engine like turbine, generator, and it is condensed back into a liquid by cold water brought up from depth and the cycle is repeated.

As we know that water is not perfectly transparent nearly all sunlight is absorbed in the surface layer which heats up. As warm water raises and cold water sinks so this warm water stays near the ocean's surface. Now Wind and waves circulate the water in the surface layer distributing the heat within it to some extend, and the temperature may remain quite uniform for the first hundred metres, but below the mixed layer the temperature drops very rapidly, perhaps 20 degrees Celsius with an additional of 150 m depth. This area of rapid transition is called thermocline and below it the temperature continues to drop with depth but very gradually.

Genrally thermocline varies with latitude and season but it is permanent in the tropics, variable in the temperate climates is strongest during the summer and is weak to nonexistent in the polar regions where the water is cold from the surface to the bottom.

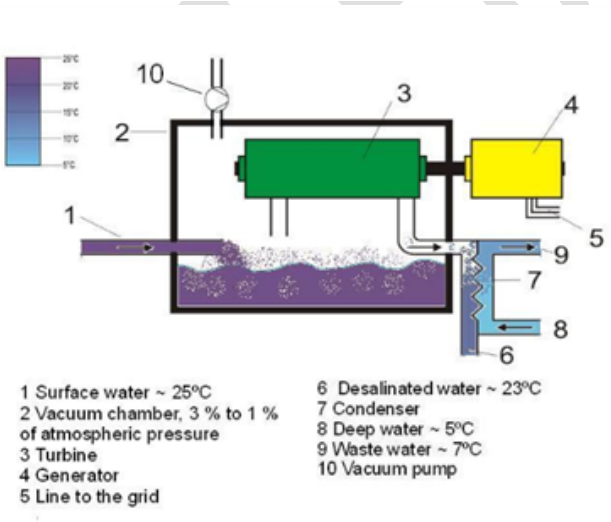


There are basically two types of OTEC power plant:



Closed Cycle OTEC system

- **Closed Cycle:** Closed cycle Ocean Thermal Energy Conversion systems use a working fluid with a low boiling point, Ammonia for example, and use it to power a turbine to generate electricity. Warm seawater is taken in from the surface of the oceans and cold water from the deep at 5°. The warm seawater vaporizes the fluid in the heat exchanger which then turns the turbines of the generator. The fluid now in the vapour state is brought in contact with cold water which turns it back into a liquid. The fluid is recycled in the system which is why it is called a closed system.



Open System OTEC system

- **Open Cycle:** Open cycle OTEC directly uses the warm water from the surface to make electricity. The warm seawater is first pumped in a low-pressure chamber where due to the drop in pressure, it undergoes a drop in boiling point as well. This causes the water to boil. This steam drives a low-pressure turbine which is attached to an electrical generator. The advantage this system has over a closed system is that, in open cycle, desalinated water in the form of steam is obtained. Since it is steam, it is free from all impurities. This water can be used for domestic, industrial or agricultural purposes.

Ocean Thermal Energy (OTEC) is a real candidate as one of the future sources of energy. Its environmental impact is negligible, in fact, the mixing of deep and shallow seawater brings up nutrients from the seafloor. The deepwater is rich in nitrates and this can also be used in agriculture.

2) State merits and demerits of OTEC plants?

Advantages:

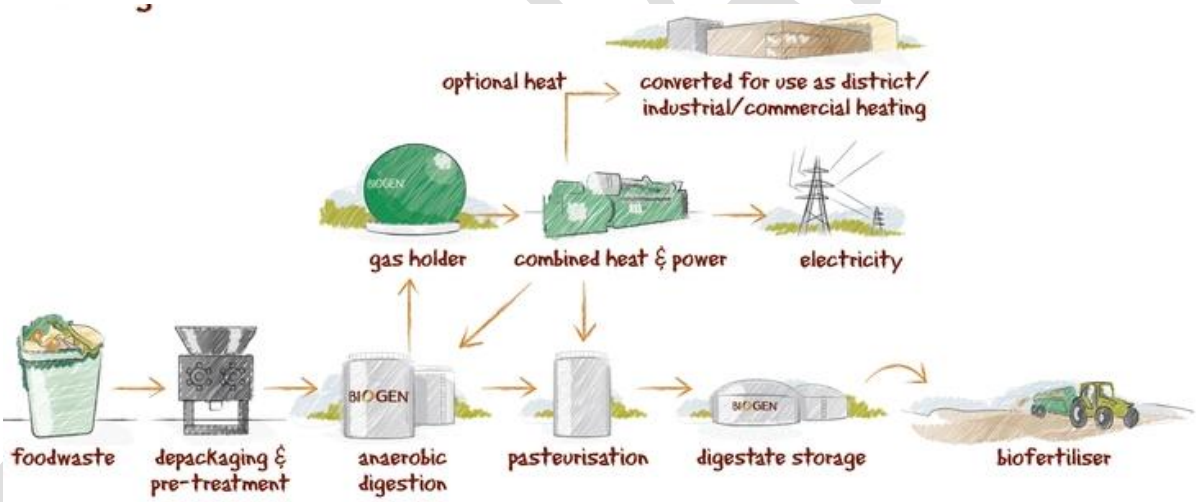
- Power developed is continuous and it is independent of weather.
- There is a small variation in power output from season to season .
- The system uses conventional power plants needing only small changes in design.
- It can produce simultaneously the desalinated water and nutrients for agriculture.

Disadvantages:

- Capital cost is very high.
- Efficiency of energy conversion is very low.
- Needs very large sized turbines due to use of low pressure of steam having high specific volume in case of open cycle.
- It uses expensive power working fluids in case of closed cycle.
- Cost of electric power generation per kWh is very high.

3) What is meant by anaerobic digestion and explain with neat sketches?

Anaerobic digestion is the process by which organic matter such as animal or food waste is broken down to produce biogas and biofertiliser. This process happens in the absence of oxygen in a sealed, oxygen-free tank called an anaerobic digester.



Anaerobic digestion is recognised by the government, Defra, the Welsh Assembly, the Scottish Parliament, Friends of the Earth and the National Farmers Union as one of the best methods for food waste recycling and dealing with farm waste and sewage sludge.

The word Anaerobic actually means ‘in the absence of oxygen’. The biogas naturally created in the sealed tanks is used as a fuel in a CHP (combined heat and power) unit to generate renewable energy i.e. electricity and heat.

What's left from the process is a nutrient rich biofertiliser which is pasteurised to kill any pathogens and then stored in large covered tanks ready to be applied twice a year on farmland in place of fossil fuel derived fertilisers.

Advantages

- Generation of biogas and fertiliser (almost complete retention of the fertiliser nutrients (N, P and K)
- Reduction of greenhouse gas emissions through methane recovery
- Combined treatment of different organic waste and wastewaters
- Reduction of solids to be handled (e.g. less excess sludge)
- Good pathogen removal depending on temperature
- Process stability (high-loads can be treated but anaerobic sludge can also be preserved for prolonged periods without any feeding)

Disadvantages

- Small- and middle-scale anaerobic technology for the treatment of solid waste in middle- and low-income countries is still relatively new
- Experts are required for the design and construction, depending on scale may also for operation and maintenance
- Reuse of produced energy (e.g. transformation into, fire/light, heat and power) needs to be established
- High sensitivity of methanogenic bacteria to a large number of chemical compounds
- Sulphurous compounds can lead to odour

4) Explain the working of Biogas plant with neat sketches?

The secret to understanding the most common biogas plant construction and working stages, is explained here for he most common Anaerobic Digestion plant type.
By far the most common biogas plants are those based upon one single anaerobic digestion tank, and they are often known as "Continuous Stirred Tank Reactors" (CSTRS).
They are often called single stage biogas plants, which can be confusing, because to the onlooker there would seem to be many more tanks and "stages".

Tanks are added to increase the amount of biogas produced, and for other uses.

Just remember that for making biogas it is the number of stages in which the methane making bugs grow to make the biogas, that count.

We will now provide an explanation of the way a biogas plant works, in 5 "steps":

Step 1 – Feeding The Bacteria That Make Biogas

Organic input materials such as silage, foodstuff remnants, fats or sludge, are fed into the biogas plant as what is called "substrate" – that's food for the biogas making bugs, to you and me!
Renewable resources such as corn, beet, or grass, can serve as feed both for animals such as cows and pigs as well as for the micro-organisms in the biogas plant.

Not only those foods work, so do farm wastes such as manure, dung, and chicken litter, plus increasingly biogas is being made from supermarket and household food waste.

Step 2 – Fermentation in the Large Tank (Digester Tank)- where the bacteria make the biogas.

In the fermenter, heated to approx. 38-40 °C, the substrate is decomposed by the micro organisms under exclusion of light and oxygen. The final product of this fermentation process is biogas with methane as the main ingredient. But, at the same time some other unwanted compounds form a small part of the biogas. Pure biogas is just methane alone. But the raw biogas that emerges can be bad smelling from corrosive hydrogen sulphide. Carbon dioxide is also produced and a percentage is contained in the biogas.

Step 3 – Fermentation Residues Storage Tank

Once the substrate has been fermented, it is pumped to the fermentation residues end storage tank and is retrieved from there for making useful products.

The residues, are called “digestate” and these comprise both a fibrous solid part and a liquid part, and these can be utilised as high quality fertiliser, and a soil conditioner respectively. The advantage of digestate is that it has a lower viscosity than farmyard manure, and it therefore penetrates into the ground more quickly than liquid manure slurry. Furthermore, the fermentation residue usually has a higher fertiliser value than manure, and is less odorous.
Further treating the fibrous part, by leaving it to mature in wind-rows, and subsequently using it as dry fertiliser is also common.

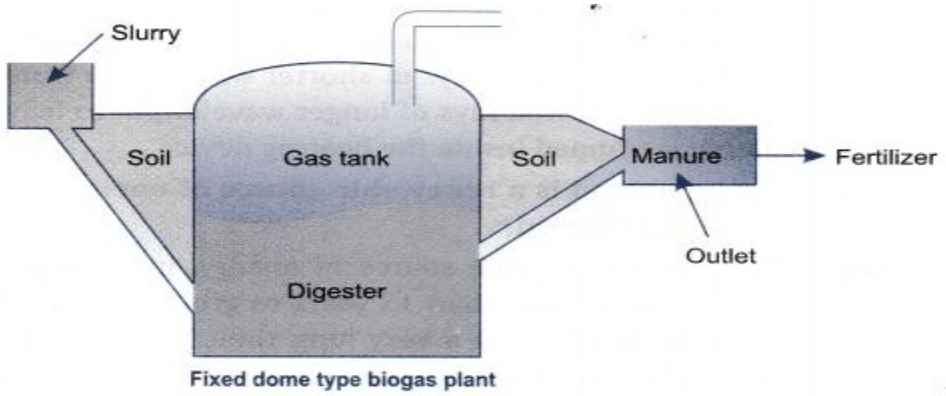
Step 4 – Using the Biogas

The biogas part of the output is usually stored in the plastic roof of the tank. From there it is most often burned in a combined heat and power (CHP) plant to generate electricity and heat.

The electric power is fed directly into the power grid.

The heat generated can be utilized to heat buildings, to dry firewood, or to dry harvest products such as grain.

At an increasing number of biogas plants the quality of the raw biogas is upgraded to purify it. The carbon dioxide, hydrogen sulphide, and other impurities are removed, and it is compressed and used as a clean renewable transport fuel.



Fixed dome type biogas plant

Step 5 – Delivery of Biogas to Users

The biogas power which is not used in running the biogas plant, is exported as either a supply to the national grid or to gas filling stations.

Every tonne of food waste recycled by anaerobic digestion as an alternative to landfill prevents between 0.5 and 1.0 tonne of CO2 entering the atmosphere, one of the many benefits of anaerobic digestion.

5) What is gasification process? List different types of gasifiers?

Gasification is a process in which combustible materials are partially oxidized or partially combusted. The product of gasification is a combustible synthesis gas, or syngas. Because gasification involves the partial, rather than complete, oxidization of the feed, gasification processes operate in an oxygen-lean environment.

Types of Gasifiers:

1. Fixed bed/moving Gasifier

Moving Bed

A diagram of a generic moving bed gasifier is shown in figure 2. Moving bed gasifiers are countercurrent flow reactors in which the coal enters at the top of the reactor and air or oxygen enters at the bottom. As the coal slowly moves down through the reactor, it is gasified and the remaining ash drops out of the bottom of the reactor. Because of the countercurrent flow arrangement, the heat of reaction from the gasification reactions serves to pre-heat the coal before it enters the gasification reaction zone. Consequently, the temperature of the syngas exiting the gasifier is significantly lower than the temperature needed for complete conversion of the coal.

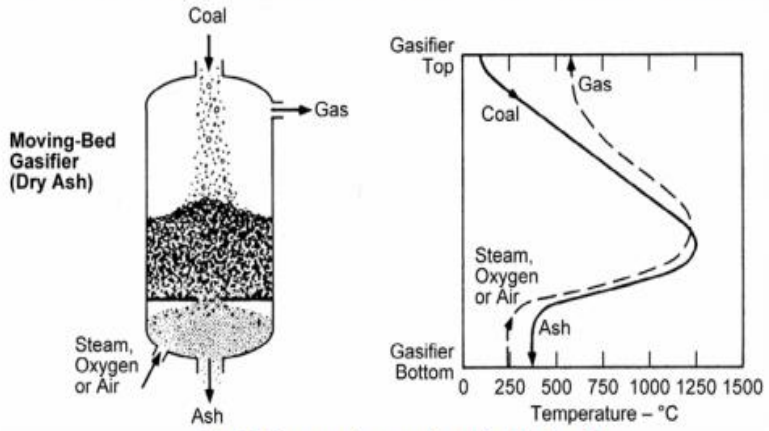


Fig. 2. Diagram of a generic moving bed gasifier

The residence time of the coal within a moving bed gasifier may be on the order of hours.

Moving bed gasifiers have the following characteristics:[1]

- Low oxidant requirements;
- Relatively high methane content in the produced gas;
- Production of hydrocarbon liquids, such as tars an oils;
- High "cold gas" thermal efficiency when the heating value of the hydrocarbon liquids are included;
- Limited ability to handle fines; and
- Special requirements for handling caking coal.

Fluidized Bed

A diagram of a generic fluidized bed gasifier is shown in figure 3. A fluidized bed gasifier is a back-mixed or well-stirred reactor in which there is a consistent mixture of new coal particles mixed in with older, partially gasified and fully gasified particles. The mixing also fosters uniform temperatures throughout the bed. The flow of gas into the reactor (oxidant, steam, recycled syngas) must be sufficient to float the coal particles within the bed but not so high as to entrained them out of the bed. However, as the particles are gasified, they will become smaller and lighter and will be entrained out of the reactor. It is also important that the temperatures within the bed are less than the initial ash fusion temperature of the coal to avoid particle agglomeration.

Typically a cyclone downstream of the gasifier will capture the larger particles that are entrained out and these particles are recycled back to the bed. Overall, the residence time of coal particles in a fluidized bed gasifier is shorter than that of a moving bed gasifier.

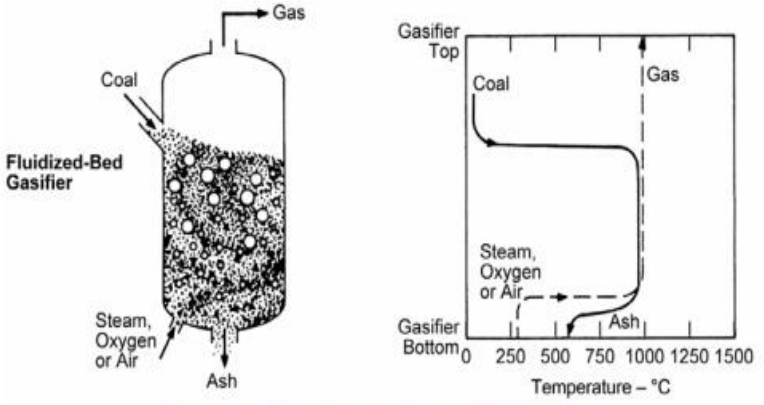


Fig. 3. Diagram of a generic fluidized bed gasifier

Generic characteristics of fluidized bed gasifiers include:[2]

- Extensive solids recycling;
- Uniform and moderate temperature; and
- Moderate oxygen and steam requirements.

6) What is Geothermal energy and explain with neat sketches?

Geothermal Energy

If you were to dig a big hole straight down into the Earth, you would notice the temperature getting warmer the deeper you go. That's because the inside of the Earth is full of heat. This heat is called geothermal energy.

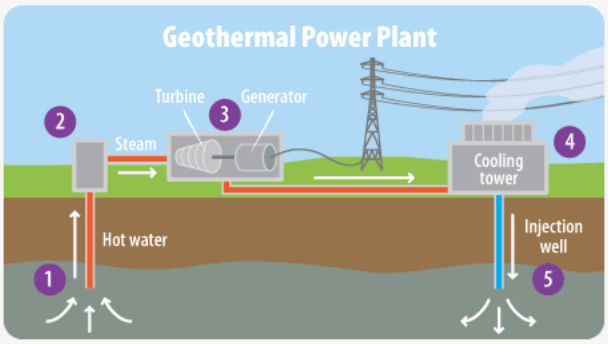
People can capture geothermal energy through:

- **Geothermal power plants,** which use heat from deep inside the Earth to generate steam to make electricity.
- **Geothermal heat pumps,** which tap into heat close to the Earth's surface to heat water or provide heat for buildings.

Geothermal Power Plants

At a geothermal power plant, wells are drilled 1 or 2 miles deep into the Earth to pump steam or hot water to the surface. You're most likely to find one of these power plants in an area that has a lot of hot springs, geysers, or volcanic activity, because these are places where the Earth is particularly hot just below the surface.

How It Works



1. Hot water is pumped from deep underground through a well under high pressure.
2. When the water reaches the surface, the pressure is dropped, which causes the water to turn into steam.
3. The steam spins a turbine, which is connected to a generator that produces electricity.
4. The steam cools off in a cooling tower and condenses back to water.
5. The cooled water is pumped back into the Earth to begin the process again.

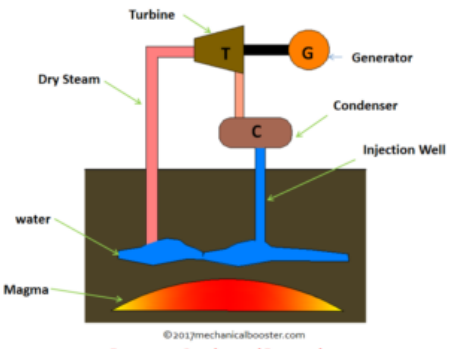
Types

The Geothermal power plant which is in working is of three types

1. Dry steam power plant
2. Flash steam power plant
3. Binary cycle power plant

1. Dry Steam Power Plant

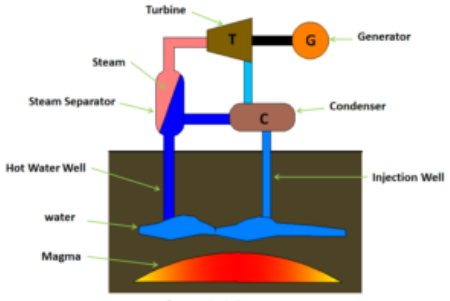
In dry steam power plant, direct steam from the geothermal reservoir is used to turn the turbine and generator to produce electricity. The temperature of the geothermal steam needed in this plant is atleast 150 degreee Celsius.



Dry steam Geothermal Power plant

2. Flash Steam Power Plant

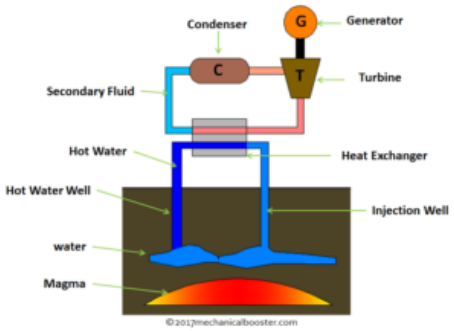
In flash steam power plant, high pressure hot water from deep inside the earth is taken out and collected in a steam separator. This high pressure hot water comes to the surface by its own and its pressure keeps on decreases as it moves upward, this allows hot water to gets converter into steam. Steam gets separated in steam separator, and allowed to turn the turbine generator. When the steam cools, it is again injected back into the earth surface to be used again. Nowadays most of the geothermal power plants used are of flash steam plants. This power plant requires a temperature of atleast 180 degree Celsius for its operation.



Flash Steam Geothermal Power plant

3. Binary Cycle Power plant

In binary cycle power plant, the heat of hot water is transferred to another liquid (called as secondary liquid). The heat of hot water causes another liquid to change into steam and then this steam is used to rotate turbine. It is the most recent developed power plant which may be operated at lowest temperature of atleast 57 degree Celsius. The secondary fluid (i.e. another liquid) used in this binary cycle geothermal power plant has much lower boiling point than water. It works on both Rankine and Kalina cycle. The thermal efficiency of this power station is expected to be lie in between 10-13%. This power plant is called as binary, since here we are using two liquids (hot water and secondary liquid) for its working.



Binary Cycle Geothermal Power plant

Advantages

1. It requires no fuel for its working.
2. It requires minimal land and fresh water.
3. Geothermal power is a sustainable source of energy.

Disadvantages

1. The fluid taken out from deep earth contains mixtures of gases such as Hydrogen sulphide (H2S), Carbon dioxide (CO2), Ammonia (NH3), methane (CH4) and Radon (Rn). If these gases is released, it will contribute to global warming, acid rain, Radiation and noxious smell.
2. It requires emission control system to reduce the exhaust that may be produce form acids and volatile chemicals.
3. The hot water taken out from the geothermal sources is hold in a solution which may contain traces of toxic chemicals like mercury, boron, arsenic, antimonny and salt. when water cools these toxic chemicals comes out of the solution and can responsible for the environmental damage if released.
4. Geothermal power plant constructed at the site may adversely affect land stability.

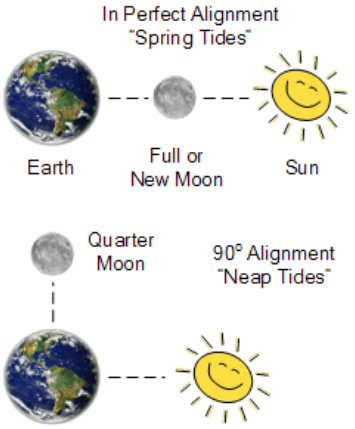
7) What is Tidal energy and explain with neat sketches?



Tidal Energy or **Tidal Power** as it is also called, is another form of hydro power that utilises large amounts of energy within the oceans tides to generate electricity. *Tidal Energy* is an "alternative energy" that can also be classed as a "renewable energy source", as the Earth uses the gravitational forces of both the moon and the sun everyday to move vast quantities of water around the oceans and seas producing tides.

As the Earth, its Moon and the Sun rotate around each other in space, the gravitational movement of the moon and the sun with respect to the earth, causes millions of gallons of water to flow around the Earth's oceans creating periodic shifts in these moving bodies of water. These vertical shifts of water are called "tides".

When the earth and the moons gravity lines up with each other, the influences of these two gravitational forces becomes very strong and causes millions of gallons of water to move or flow towards the shore creating a "high tide" condition. Likewise when the earth and the moons gravity are at 90° to each other, the influences of these two gravitational forces is weaker and the water flows away from the shore as the mass of water moves to another location on the earth, creating a "low tide" condition. This ebbing and flowing of the tides happens twice during each period of rotation of the earth with stronger weekly and annual lunar cycles superimposed onto these tides.



Alignment of the Moon and Sun on Tides

When the moon is in perfect alignment with the earth and the sun, the gravitational pull of the moon and sun together becomes much stronger than normal with the high tides becoming very high and the low tides becoming very low during each tidal cycle. Such tides are known as *spring tides* (maximum). These spring tides occur during the full or new moon phase.

The other tidal situation arises during *neap tides* (minimum) when the gravitational pull of the moon and the sun are against each other, thus cancelling their effects. The net result is a smaller pulling action on the sea water creating much smaller differences between the high and low tides thereby producing very weak tides. Neap tides occur during the quarter moon phase. Then spring tides and neap tides produce different amounts of potential energy in the movement of the sea water as their effects differ from the regular high and low sea levels and we can use these tidal changes to produce renewable energy. So we can say that the tides are turning for alternative energy.

So we now know that the constant rotational movement of the earth and the moon with regards to each other causes huge amounts of water to move around the earth as the tides go in and out. These tides are predictable and regular resulting in two high tides and two low tides each day with the level of the oceans constantly moving between a high tide and a low tide, and then back to a high tide again. The time taken for a tidal cycle to happen is about 12 hours and 24 minutes (called the "diurnal cycle") between two consecutive high tides allowing Oceanographers and Meteorologist to accurately predict the ebb and flow of the tides around the oceans many years in advance.

The main big advantage of this is that the tides are therefore perfectly predictable and regular unlike wind energy or solar energy, allowing miles of coastline to be used for tidal energy exploitation and the larger the tidal influence, the greater the movement of the tidal water and therefore the more potential energy that can be harvested for power generation. Therefore Tidal Energy can be considered as a renewable energy source as the oceans energy is replenished by the sun as well as through tidal influences of the moon and suns gravitational forces.

Tidal Energy Generation

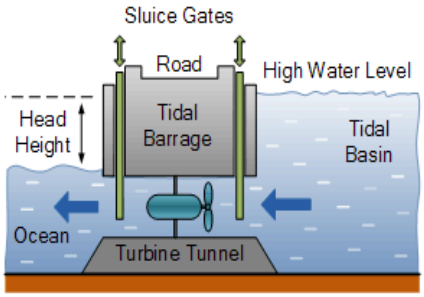
Since the position of the earth and the moon with respect to the sun changes throughout the year, we can utilise the potential energy of the water contained in the daily movement of the rising and falling sea levels to generate electricity. The generation of electricity from tides is similar in many ways to hydro-electric generation we looked at in the hydro energy tutorials. The difference this time is that the water flows in and out of the turbines in both directions instead of in just one forward direction.

Tidal energy, just like hydro energy transforms water in motion into a clean energy. The motion of the tidal water, driven by the pull of gravity, contains large amounts of kinetic energy in the form of strong tidal currents called tidal streams. The daily ebbing and flowing, back and forth of the oceans

tides along a coastline and into and out of small inlets, bays or coastal basins, is little different to the water flowing down a river or stream.

The movement of the sea water is harnessed in a similar way using waterwheels and turbines to that used to generate hydro electricity. But because the sea water can flow in both directions in a tidal energy system, it can generate power when the water is flowing in and also when it is ebbing out. Therefore, tidal generators are designed to produce power when the rotor blades are turning in either direction. However, the cost of reversible electrical generators are more expensive than single direction generators.

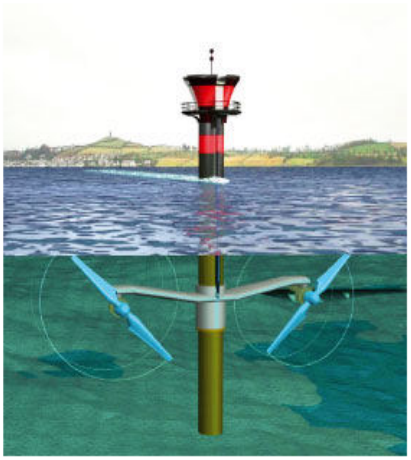
Different Types of Tidal Energy Systems



Tidal Barrage – A *Tidal Barrage* is a type of tidal power generation that involves the construction of a fairly low dam wall, known as a "barrage" and hence its name, across the entrance of a tidal inlet or basin creating a tidal reservoir. This dam has a number of underwater tunnels cut into its width allowing sea water to flow through them in a controllable way using "sluice gates". Fixed within the tunnels are huge water turbine generators that spin as the water rushes past them generating tidal electricity.

Tidal barrages generate electricity using the difference in the vertical height between the incoming high tides and the outgoing low tides. As the tide ebbs and flows, sea water is allowed to flow in or out of the reservoir through a one way underwater tunnel system. This flow of tidal water back and forth causes the water turbine generators located within the tunnels to rotate producing tidal energy with special generators used to produce electricity on both the incoming and the outgoing tides.

The one disadvantage of **Tidal Barrage Generation**, is that it can only generate electricity when the tide is actually flowing either "in" or "out" as during high and low tide times the tidal water is stationary. However, because tides are totally predictable, other power stations can compensate for this stationary period when there is no tidal energy being produced. Another disadvantage of a tidal barrage system, is the environmental and ecological effects that a long concrete dam may have on the estuaries they span.



Tidal Stream – A *Tidal Stream Generation* system reduces some of the environmental effects of tidal barrages by using turbine generators beneath the surface of the water. Major tidal flows and ocean currents, like the Gulf Stream, can be exploited to extract its tidal energy using underwater rotors and turbines.

Tidal stream generation is very similar in principal to wind power generation, except this time water currents flow across a turbines rotor blades which rotates the turbine, much like how wind currents turn the blades for wind power turbines. In fact, tidal stream generation areas on the sea bed can look just like underwater wind farms.

Unlike off-shore wind power which can suffer from storms or heavy sea damage, tidal stream turbines operate just below the sea surface or are fixed to the sea bed. Tidal streams are formed by the horizontal fast flowing volumes of water caused by the ebb and flow of the tide as the profile of the sea bed causes the water to speed up as it approaches the shoreline.

As water is much more denser than air and has a much slower flow rate, tidal stream turbines have much smaller diamet and higher tip speed rates compared to an equivalent wind turbine. Tidal stream turbines generate tidal power on both ebb and flow of the tide. One of the disadvantages of **Tidal Stream Generation** is that as the turbines are submerged un the surface of the water they can create hazards to navigation and shipping.

Other forms of tidal energy include tidal fences which use individual vertical-axis turbines that are mounted within a fer structure, known as the caisson, which completely blocks a channel and force water through them. Another alternative way harnessing tidal power is by using an "oscillating tidal turbine". This is basically a fixed wing called a **Hydroplane** positior on the sea bed. The hydroplane uses the energy of the tidal stream flowing past it to oscillate its giant wing, similar t whales flipper, up and down with the movement of the tidal currents. This motion is then used to generate electricity. T angle of the hydroplane to the flow of the tide can be varied to increase efficiency.

Tidal energy is another form of low-head hydro power that is completely carbon neutral like wind and hydro energy. Ti power has many advantages compared to other forms of renewable energy with its main advantage being that it predictable. However, like many other forms of renewable energy, tidal energy also has its disadvantages such as inflexible generation times dependant upon the tides and the fact that it operates in the hostile conditions of the oceans a seas. So here are some of the advantages and disadvantages associated with "tidal energy".

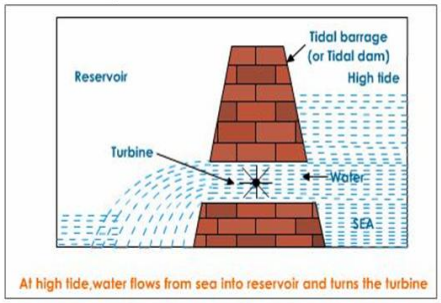
Advantages of Tidal Energy

- Tidal energy is a renewable energy resource because the energy it produces is free and clean as no fuel is needed and no waste bi-products are produced.
- Tidal energy has the potential to produce a great deal of free and green energy.
- Tidal energy is not expensive to operate and maintain compared to other forms of renewable energies.
- Low visual impact as the tidal turbines are mainly if not totally submerged beneath the water.
- Low noise pollution as any sound generated is transmitted through the water.
- High predictability as high and low tides can be predicted years in advance, unlike wind.
- Tidal barrages provide protection against flooding and land damage.
- Large tidal reservoirs have multiple uses and can create recreational lakes and areas where before there were none.

Disadvantages of Tidal Energy

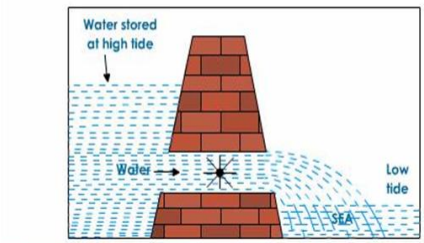
- Tidal energy is not always a constant energy source as it depends on the strength and flow of the tides which themselves are effected by the gravitational effects of the moon and the sun.
- Tidal Energy requires a suitable site, where the tides and tidal streams are consistently strong.
- Must be able to withstand forces of nature resulting in high capital, construction and maintenance costs.
- High power distribution costs to send the generated power from the submerged devices to the land using long underwater cables.

During high tide, when the level of water in the sea is high, sea-water flows into the reservoir of the barrage and turns the turbines. The turbines then turn the generator shaft to produce electricity.



At high tide,water flows from sea into reservoir and turns the turbine

During low tide, the sea-water stored in the barrage reservoir is allowed to flow out into the sea. This flowing water also turns the turbines and generates electricity. Thus, as the sea-water flows in and out of the tidal barrage during high and low tides, the turbines rotate continuously to generate electricity.



At low tide, stored water flows out from reservoir into sea and turns the turbine

8) What is Wave energy and explain with neat sketches?

Wave Energy also known as **Ocean Wave Energy**, is another type of ocean based renewable energy source that uses the power of the waves to generate electricity. Unlike tidal energy which uses the ebb and flow of the tides, *wave energy* uses the vertical movement of the surface water that produce tidal waves. Wave power converts the periodic up-and-down movement of the oceans waves into electricity by placing equipment on the surface of the oceans that captures the energy produced by the wave movement and converts this mechanical energy into electrical power.

- The wave energy is developed due to the winds interacting with the surface of the ocean. The wind gets about 1% to 5% of sun's energy and part of this energy is transferred to the waves.
- The availability of this wave energy depends upon the height and frequency of waves. It is estimated that about 10 kw of - energy is available for 1m of wave front and its potential in India is estimated to 40000mw. The power developed can be estimated by

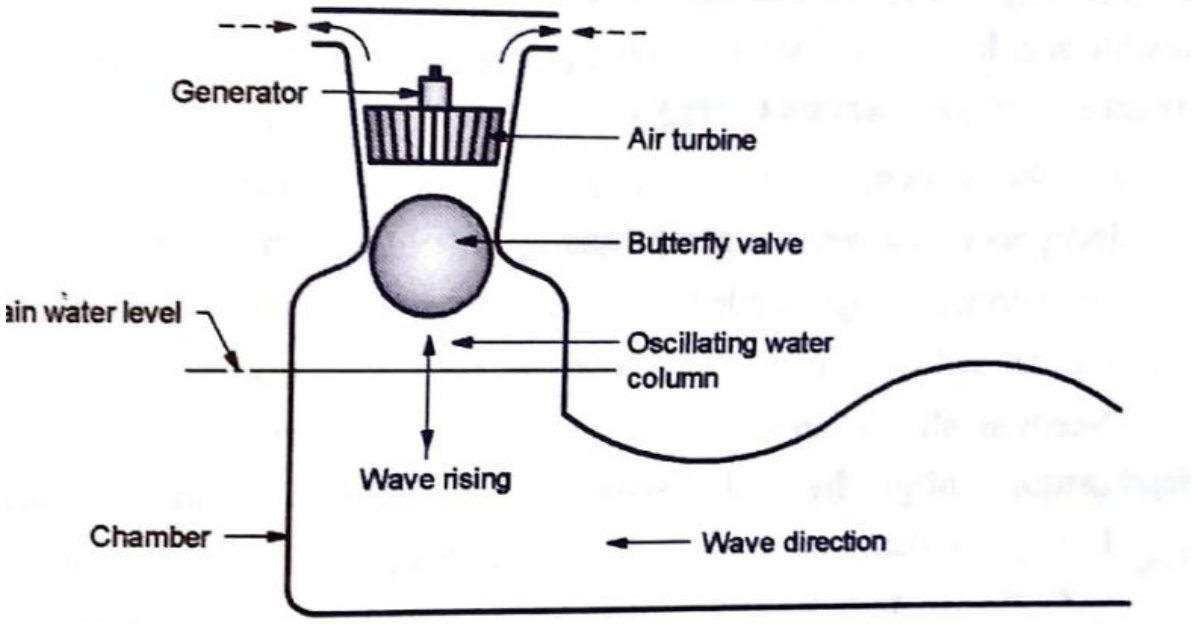
  Power , P = 0.55H2 . t

  Where,

  H = Wave Height

  T = time period in seconds

  P = power in kw.
- In India, 150 kw system utilizing wave energy has been installed at Thiruvananthapuram.
- The most common wave energy conversion system is the Oscillation water column system .



- It consists of a chamber in the sea exposed to wave action through an opening on the side. It compresses the air column above it.
- Compressed air flow through the butterfly valve and runs the air turbine connected to the induction generator which produces electric power.
- When the waves recede, it causes suction of air and the atmosphere air is drawn in to the chamber. Therefore, these is a regular oscillatory movement of water column causing to and fro of air motion which drives the air turbine.

Wave Energy Advantages

- Wave energy is an abundant and renewable energy resource as the waves are generated by the wind.
- Pollution free as wave energy generates little or no pollution to the environment compared to other green energies.
- Reduces dependency on fossil fuels as wave energy consumes no fossil fuels during operation.
- Wave energy is relatively consistent and predictable as waves can be accurately forecast several days in advance.
- Wave energy devices are modular and easily sited with additional wave energy devices added as needed.
- Dissipates the waves energy protecting the shoreline from coastal erosion.
- Presents no barriers or difficulty to migrating fish and aquatic animals.

Wave Energy Disadvantages

- Visual impact of wave energy conversion devices on the shoreline and offshore floating buoys or platforms.
- Wave energy conversion devices are location dependent requiring suitable sites were the waves are consistently strong.
- Intermittent power generation as the waves come in intervals and does not generate power during calm periods.
- Offshore wave energy devices can be a threat to navigation that cannot see or detect them by radar.
- High power distribution costs to send the generated power from offshore devices to the land using long underwater cables.
- They must be able to withstand forces of nature resulting in high capital, construction and maintenance costs.